Credit Risk Research - Findings and Gap (11 Papers)

Overall Finding (My Thesis)

- Read abstracts of 11 research papers on credit risk in banking (Chinese banks, Indian banks, and foreign/generic datasets).
- Most papers split into two groups that never combine with each other: theory-only papers, and model-building papers with old or foreign data.
- Three papers touch India directly, but each is incomplete in a different way (explained below).
- No paper combines real, recent Indian bank data with a modern prediction model and an actionable, explainable output for the bank.
- This is the core gap our project can target.

Group A: Theory Only Papers (No Data, No Model)

- These papers only explain why credit risk happens or how banks manage it in general.
- No real data used, no prediction model built or tested.
- Mostly China focused, mostly old (2005 to 2011).

Group B: Model Building Papers (But With Problems)

- These papers do build machine learning or statistical models, but each has a different problem:
    - Wrong or confused method: one paper mixes up two different algorithms as if they are the same.
    - Old data: some models are trained on data from 2005.
    - Foreign or generic data, not Indian: models built on Chinese, Taiwanese, or Kaggle/German credit datasets.
    - Very small sample size: some models trained on only 70 to 243 records, or just 1,000 generic records.
    - Only basic prediction: most models only say default or no default, nothing actionable for the bank, and none explain why the model decided what it decided.

The Three India-Specific Papers (Each Incomplete in a Different Way)

- Paper on Indian private banks (2025): only surveys employee opinions, not real loan or default data. Admits loan-level data studies in India are still missing.
- Paper using real Axis Bank loan data (2007-2015) with Random Forest and XGBoost: this is the closest paper to our gap so far, since it does use real Indian bank data with modern ML.
    - But it only predicts default versus non-default, and forecasts trends. It does not sort customers into risk segments the bank can act on (like different interest rates or credit limits).
    - It only mentions explainability tools (SHAP, LIME) as a future idea. It never actually builds an explainable model.
    - Its data stops at 2015 and its forecasts only go up to 2020. It does not reflect current Indian retail credit stress or recent RBI digital lending and co-lending rules.
- Paper studying how 8 major Indian banks actually use AI (from their annual reports, 2024): this is not a model-building paper, it is a study of real bank practice.
    - Finds that Indian banks use AI mostly for retail credit scoring and fraud detection, not for full credit risk evaluation across their loan book.
    - Finds banks do not disclose AI usage consistently, and there is no regulatory framework or certification process for validating these AI credit models.
    - This supports our gap from the real-world side: even in practice, Indian banks have not built the kind of actionable, explainable credit risk model we are proposing.

Final Research Gap

- No existing paper or bank practice combines all of the following together:
    - Real, recent Indian private bank data (actual loans and customers, not survey opinions, and not stopping in 2015)
    - A modern, well-tested prediction model built for this data
    - An actionable output for the bank, such as risk segments or categories, not just default or no default
    - An explainability layer (such as SHAP or LIME) that is actually built and tested, not just mentioned as a future idea
- Extra angle nobody has touched: India-specific factors such as RBI co-lending and digital lending rules, rising stress in personal loans and credit cards, and newer data sources like digital transaction history.
- This is the research gap: build a credit risk prediction model using real, recent Indian private bank data, using modern ML methods, that gives the bank an actionable and explainable output such as risk categories, something no existing paper or current bank practice has fully done for the Indian market.